

CSCI E-25, Computer Vision

Project proposal:

Document Image Classification - Multiclass classifier

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Introduction:

In digital libraries, documents are often stored as images before they are processed by text extraction tools. Before extracting text from the image, classification of a document is an important step for faster and relevant information retrieval as well as record-keeping. The document image classification covers different types of documents like letter, form, email etc.

Content-based analysis of document images has a number of applications. The economic feasibility of creating a large database of document images has left a tremendous need for robust ways to access the information. Printed documents are scanned for archiving or in an attempt to move towards a paperless office and are stored as images. Documents play a pivotal role in all of the fields of business communication and record-keeping.

Complex documents present a great challenge to the field of document recognition and retrieval. The primary task of processing these complex documents is to isolate the different contents present in the documents.

Project goal:

Build a model to classify input document image as one of sixteen predefined categories. The project will use publicly available dataset, apply preprocessing, feature extractions, and modeling to categorize the image.

Dataset:

The RVL-CDIP (Ryerson Vision Lab Complex Document Information Processing) dataset consists of 400,000 grayscale images in 16 classes, with 25,000 images per class. There are 320,000 training images, 40,000 validation images, and 40,000 test images. The images are sized so their largest dimension does not exceed 1000 pixels.

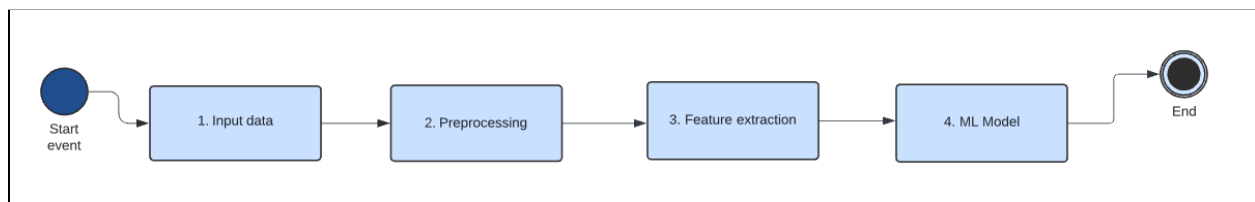
Dataset size is 37GB (compressed), available at rvl-cdip.tar.gz. The categories are numbered 0 to 15, in the following order: 0) letter 1) form 2) email 3) handwritten 4) advertisement 5) scientific report 6) scientific publication 7) specification 8) file folder 9) news article 10) budget 11) invoice 12) presentation 13) questionnaire 14) resume, and 15) memo.

The compressed file contains a directory structure as shown below. Labels directory contains files for training, test and validation data. These files have an image name, along with an associated category as space separated data.

Name	Date Modified	Size
▼ rvl-cdip	Yesterday at 1:25 PM	--
readme.txt	Jun 5, 2015 at 4:55 PM	2 KB
▼ labels	Yesterday at 2:11 PM	--
val.txt	Dec 31, 2014 at 3:37 PM	1.7 MB
train.txt	Dec 31, 2014 at 3:37 PM	13.7 MB
test.txt	Dec 31, 2014 at 3:37 PM	1.7 MB
> images	Yesterday at 1:25 PM	--

Computer vision pipeline:

Following diagram shows high-level methodology to address the problem statement.



Infrastructure details:

I am planning to use Google's colab, with google's drive as storage. For this project, I will start with Colab Pro, but possibly be required to upgrade my subscription for Colab Pro+. This will allow the use of GPUs for faster processing during the exploratory phase. Also, starting Google drive's subscription for 100GB storage, which should be enough for a given dataset.

Input data:

RVL-CPID dataset has a large number of images for training, testing, and validating the model. It may require higher processing power with larger memory, and storage. The dataset size is around 50GB (extracted). As shown above, images are stored differently than labels, with mapping files.

In this step, images will be sorted, rearranged such that the same category images are under the same folder. This will allow us to create a smaller dataset with all categorized having equal number of images. This will avoid imbalance classification problems.

Outcome of this step will be easy to run jupyter notebook which will take following parameters to generate smaller dataset:

- Number of train images
- Number of test images
- Number of validation images
- Output path

Note: This step will simply rearrange the images, generate a smaller dataset, along with a mapping file similar to the original dataset. For this project, I am planning to use only training and test data. Test data will be used to measure model's performance on unseen data. The use of this step is for ground-up models with smaller dataset, before generalizing to full dataset.

Preprocessing:

RVL-CPID images are mostly 1000 x 754, but not all. There are many images with different dimensions. During the preprocessing step, I will resize images to 512 x 512 pixels. As resizing of images will result in downsizing of original images, I have to consider the "anti-aliasing" effect to be addressed.

I will resize only those images, which are identified in the input data step. Resizing of images is required to unify images with the same size, which will help in subsequent steps. Also, downsizing of original images to 512 x 512 pixels, makes 262, 144 features in 1-D arrays.

Outcome of this step will be a dataset ready for feature extraction, where all images are 512 x 512 pixels.

Feature extraction:

Once images are resized, I will perform a few feature extraction steps to explore possibilities of further image size reduction. This will include creating a 1-D flattened array of trained images to display the feature matrix.

As this project is about handling document images, it is possible that different types of documents may be derived based on regional features like header, footer, left and right margins, and document body. For that, images can be split into smaller regions, run the mentioned pipeline for each region individually, and the final result will be derived based on aggregation results. Region extraction is aspirational at this point for this project, due to limitation of time, and computing, I may not explore this.

I will use Convolution Neural Net (CNN) for feature extraction. I will compare feature extraction with different numbers of channels (3, 5, and 7), with different types of weight calculations like parameter sharing, Pooling and Invariance, as well as transfer learning from pre-trained networks.

Outcome of this step will be a columnar dataset where every row represents an image, while every column will be the result of multiple convolution processes.

ML Model:

Once convolution layers are converted to 1-D features, we can apply different classifiers for final results. I am planning to explore different classifiers like NeuralNet with different hidden layers, activation functions as well as number of epochs. Also, aspirationally, I would like to explore Support Vector Machine (SVM), as well as Adaboosting classifier..

Outcome of this step will be the model's training accuracy, recall as well as precision matrix. I am also planning to have a performance matrix representing runtime for each model with different hyper-parameters.. This matrix will allow us to select the best model for the given problem statement.

Conclusion:

As a final step of the modeling process, I will validate different models against test data. The comparison results will be shown as graphs, tabular format along with recommendation.

Reference:

Dataset <https://www.kaggle.com/pdavpoojan/the-rvldip-dataset-test>

RVL-CDIP(Ryerson Vision Lab Complex Document Information Processing)
<https://medium.com/analytics-vidhya/rvl-cdip-ryerson-vision-lab-complex-document-information-processing-aa30b00a2b1e>

Evaluation of Deep Convolutional Nets for Document Image Classification and Retrieval
<https://arxiv.org/pdf/1502.07058v1.pdf>